Assignment IV Report

Perception from Sensor Data for Robot Manipulation

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1 Perception Pipeline (`perception.py`)

1.1 Boustrophedon Exploration Strategy

The `generate_exploration_poses` function generates a structured sweep of the scene using a boustrophedon (snake-path) traversal over a grid of yaw and pitch angles. Three pitch levels (`-0.9, -0.6, -0.3`) are swept, and for each level the list of five yaw angles $(-\pi/2, -\pi/4, 0, \pi/4, \pi/2)$ is reversed in alternating rows. This avoids redundant return movements. Starting from the robot home configuration, joint angles 0 and 1 are varied while keeping the rest at their home values, covering the full frontal hemisphere of the workspace.

1.2 Offset Camera Approach for Object Identification

In `identify_objects`, the robot approaches each cluster using a lateral XY offset $(\Delta x, \Delta y) = (-0.1, -0.1)$ rather than a directly overhead approach. The end effector is moved to two successive Z-heights (0.42 m and 0.38 m above the cluster centroid) with intermediate pauses of 0.5 s each to allow camera settling. This avoids the camera being directly above the object, which can cause the object to appear visually distorted or partially occluded by the gripper in the image frame.

The YOLO detector is run on the captured RGB frame. The 3D cluster centroid is projected into the image using the pinhole model (via `project_world_to_image`) and checked against predicted bounding boxes. The closest bounding box center to the projected cluster centroid is selected as the matched label.

An additional near-origin filter (`np.linalg.norm(center) < 0.15`) discards degenerate clusters corresponding to the robot base or other artifacts near the origin.

1.3 Purely Geometric Pick Pose Estimation

`get_object_pick_pose` computes the grasp target entirely from 3D cluster geometry. The XY centroid is taken from the axis-aligned bounding box center, but the Z coordinate is replaced with the maximum Z bound of the bounding box — i.e., the top surface of the object. This prevents the gripper from attempting to reach the volumetric centroid, which may be below the surface, and instead targets the topmost extent suitable for a top-down grasp.

1.4 Z-Clearance Offset for Placement

get_obj_place_pose places objects at the maximum Z bound of the target cluster plus an additional 0.05 m offset. This clearance prevents the robot from attempting to descend into already-occupied plate space when stacking, ensuring the release position is above the top surface of any previously placed objects.

1.5 Iterative RANSAC Plane Removal

remove_planes applies two rounds of RANSAC-based plane segmentation. This double removal targets the ground surface in the first pass and the table top in the second pass.

2 Motion Control (control.py)

2.1 Bounded PD Position Control with Convergence Guard

move_joints commands joint angles using setJointMotorControlArray in position control mode. The positionGains parameter is explicitly set to 0.03 per joint, replacing the default high-gain behavior with a lower-gain, slower but smoother interpolation. Rather than waiting indefinitely for convergence, a hard cap of 500 simulation steps is applied before the function exits regardless of motion state, preventing the pipeline from stalling when a joint cannot reach its target. A 0.5 s real-time sleep after the motion loop adds additional settling time.

3 Task Planning (planning.py)

3.1 Size-Based Filtering to Exclude ungraspable objects

Before adding a pick-and-place action for an object, get_clear_task_plan computes the minimal oriented bounding box (MOBB) of each cluster and checks whether its smallest dimension is less than 0.1 m. Objects with all three dimensions above this threshold (such as the shelf) are excluded. Oriented bounding boxes are used rather than axis-aligned ones specifically to handle non-axis-aligned objects more accurately. Small, compact clusters (fruits) pass this test and are added to the plan.

4 Contribution

This project was completed with equal contribution from both students. Each student was involved in the design, implementation, experimentation, analysis of results, and preparation of the report. The work distribution was balanced throughout all stages of the project (50–50 contribution).